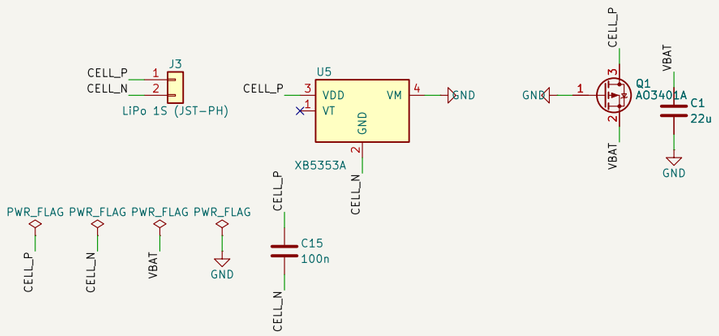
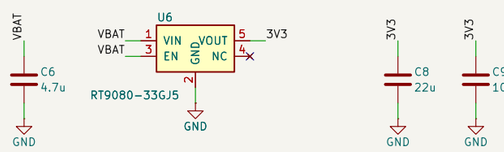
Battery, protection, reverse polarity



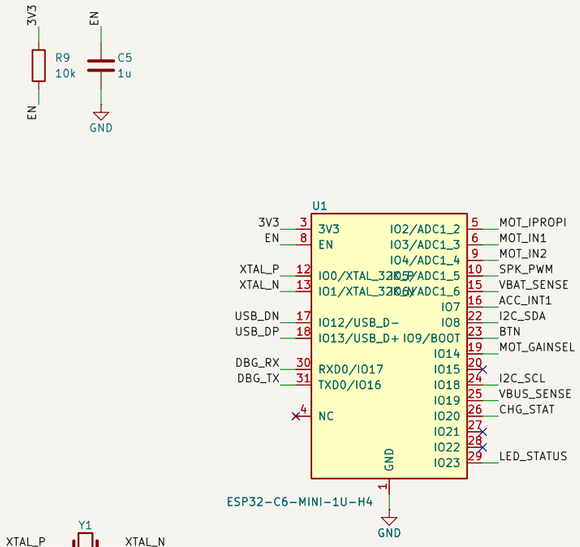
U5: VM (pins 4,5) is pack minus = board GND. Keep CELL_N –> U5 –> GND short and wide.
Q1: body diode conducts until U5 closes, then the channel turns fully on (Vgs = –Vcell).
Reversed cell: Q1 blocks the high side, U5 blocks the low side.

3.3 V regulator



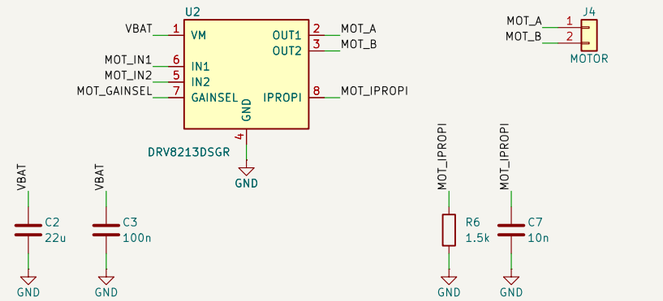
2 uA quiescent, 600 mA, 0.31 V dropout at 600 mA. C8 bulk at the regulator, C9 at module pin 3.

ESP32-C6 module (Zigbee / Thread / BLE / Wi-Fi, USB)



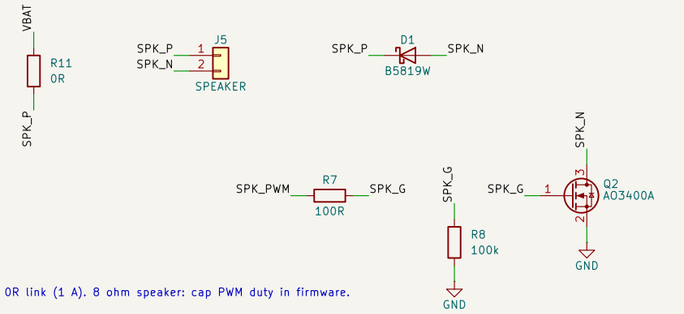
EN: 10k / 1u power-on delay (Espressif). IO9 = BOOT (SW1). IO8 strap high via R16.
IO0/IO1: 32.768 kHz crystal for the light-sleep RTC. IO12/13: USB Serial/JTAG on the charging port.
ADC1: IO2 MOT_IPROPI, IO6 VBAT_SENSE. LP (deep-sleep wake): IO7 ACC_INT1.

Motor driver (DRV8213, current sense)



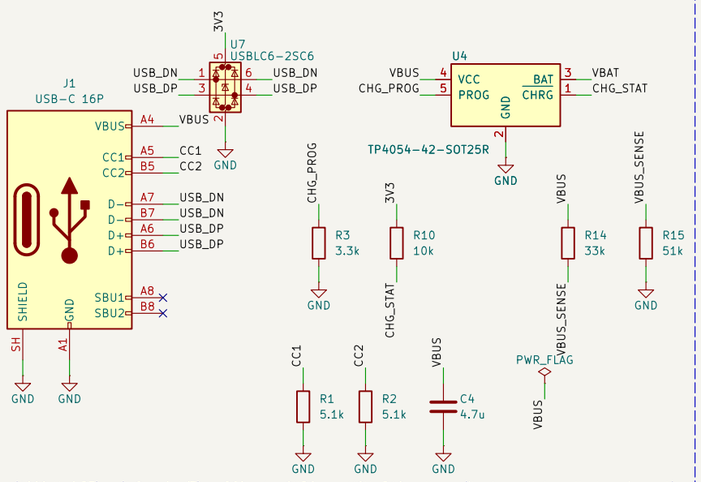
I_limit = 0.51 V / (R6 x A_IPROPI); GAINSEL low 1.66 A, Hi-Z 324 mA, high 69 mA.
C7: IPROPI filter (datasheet 8.4.2.2). 15 us with R6, against motor-cap spikes.
Auto-sleep (20 nA) when IN1 = IN2 = 0. Keep the 100 nF cap on the motor tabs.

Speaker (optional, PWM)



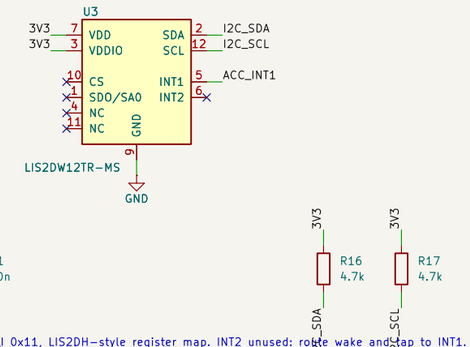
R11: OR link (1 A). 8 ohm speaker: cap PWM duty in firmware.

USB-C (sink), LiPo charger, USB detect



I_chg = (1000 V / R3) x (1.2 – 4 I/3) = 260 mA. CHRG -> MCU, R10 pull-up (its done-state pull-down is weak).
U7 pin 5 on 3V3, not VBUS: the C6 pulls D+ up while unplugged and would back-feed VBUS.
VBUS_SENSE = 0.61 x VBUS (3.0 V at 5 V): plugged in, charging or full.

Accelerometer (wake-on-motion)



MSKSEMI part, not ST: WHO_AM_I 0x11, LIS2DH-style register map. INT2 unused: route wake and tap to INT1.
CS and SDO/SA0 open (internal pull-ups): I2C, address 0x19. C10 at VDD, C11 at VDDIO.

Notes

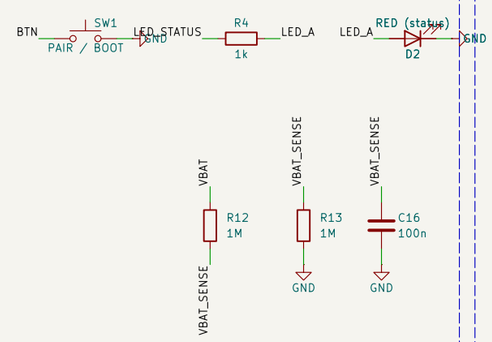
One board for all three toys (bird / fish / ball). 1S LiPo, protected or not.

Battery voltage: VBAT_SENSE (IO6, ADC1_CH6), VBAT / 2.
On charger: VBUS_SENSE (IO19). Charging: CHG_STAT (IO20), low = charging.
Motor current: MOT_IPROPI (IO2, ADC1_CH2), 0 dB attenuation.
GAINSEL: drive low / drive high / input-disconnected = Hi-Z.
Motor PWM = drive/brake: fwd IN1=1, IN2=1-duty; rev IN2=1, IN1=1-duty.
Motor: IN1 IO3, IN2 IO4, GAINSEL IO14. Speaker: SPK_PWM IO5. Status LED: LED_STATUS IO23.

Test points: value = back-silk label. TX/RX = UART0, 3.3 V. BOOT low + EN pulse = download mode.
Hand-soldered, not JLC-assembled: J3 battery (JST-PH, the plug cells come with).
Antenna: 2.4 GHz FPC antenna on the module u.FL, plugged in by hand.
H1, H2: M2 mounting holes (NPTH), diagonal corners; ISO 7380 button head (3.5 mm).

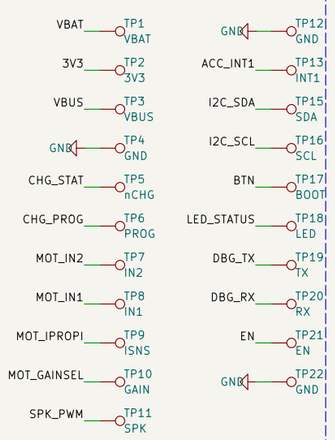
H1 M2
H2 M2

Button, status LED, battery sense



D2: GPIO high = on, 1.3 mA.
VBAT_SENSE = VBAT / 2, 2.1 uA. C16 holds it for the ADC sample.

Test points (back)



One board for bird / fish / ball – see README.md
ESP32-C6-MINI-1U + RT9080 + DRV8213 + LIS2DW12TR-MS + TP4054 + XB5353A, 1S LiPo

zigbee_toy

Sheet: /
File: zigbee_toy.kicad_sch

Title: Zigbee cat toy controller

Size: A3	Date: 2026-10-20	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/1